Seeing Motion in Data

Applying Image Formation Theory to Human Activity Recognition

Chakilam Lalith Sai Velagapudi Venkata Akhil Eslavath Ajaynayak

Indian Institute of Technology Roorkee
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Outline

1. Motivation & Problem Setting
2. Theoretical Framework
3. Experimental Setup
4. Results & Analysis
5. Conclusion & Future Work

The Challenge of Inertial Signals

- **Human Activity Recognition (HAR)** relies on smartphone accelerometers and gyroscopes (50 Hz).
- Raw signals are noisy, drift between subjects, and have long temporal dependencies.
- Traditional approaches use hand-crafted features, which require domain expertise and generalize poorly.

The Image Formation Hypothesis

The Core Idea

If we can mathematically transform 1D multi-channel time-series data into 2D images, we can utilize the powerful spatial pattern recognition capabilities of Convolutional Neural Networks (CNNs).

Inspiration: Liu et al. (2021) demonstrated 96.09% accuracy on a photovoltaic fault-detection dataset using an image formation pipeline.

Our Project's Question: Does this theory successfully transfer to HAR, which operates in a physically different domain with fewer channels?

The Image Formation Pipeline

The mathematical framework applies three operators in sequence:

$$\tilde{\gamma} = \gamma \circ \beta \circ \alpha$$

1. α **Operator (Signal Ordering):** Gathers signals into a prescribed sequence.
2. β **Operator (Normalization):** Standardizes amplitudes.
3. γ **Operator (Spatial Mapping):** Positions signals into a pixel grid.

Step 1 & 2: Ordering (α) and Normalization (β)

The α Operator: Sorts the 9 available channels into specific indices:

- Channels 0–2: Body Acceleration (x, y, z)
- Channels 3–5: Body Gyroscope (x, y, z)
- Channels 6–8: Total Acceleration (x, y, z)

The β Operator: Applies channel-wise min-max normalization based on training data to map values to pixel intensities $[0, 1]$:

$$\hat{x}_i(k) = \frac{x_i^{raw}(k) - \min_a x_i^{raw}(a)}{\max_b x_i^{raw}(b) - \min_a x_i^{raw}(a)}$$

Step 3: Spatial Mapping (γ) and Rigidity

The γ Operator: At each time step k, maps the 9 normalized values to a 3×3 pixel patch:

$$\gamma(k) : x(k) \mapsto y(k) \in [0, 1]^{3 \times 3}$$

Sequential Image Formation:

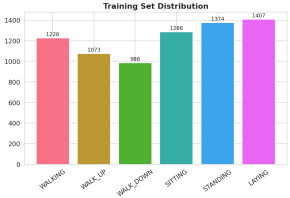
- A full 2.56-second window has 128 time steps $\rightarrow$ 128 patches.
- Patches are tiled into an 8×16 grid.
- **Final Image Size:** $(8 \times 3) \times (16 \times 3) = 24 \times 48$ pixels.

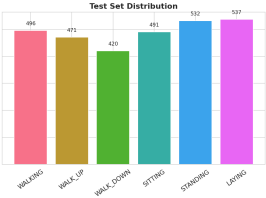
Rigidity Property

This is a deterministic, lossless transformation. The same raw signal always produces the exact same image, allowing a CNN to generalize predictably.

The UCI HAR Dataset

- **Subjects:** 30 volunteers (aged 19–48).
- **Activities:** 6 classes (Walking, Upstairs, Downstairs, Sitting, Standing, Laying).
- **Windows:** 2.56s sliding windows with 50% overlap.
- **Split:** 7,352 Train / 2,947 Test.



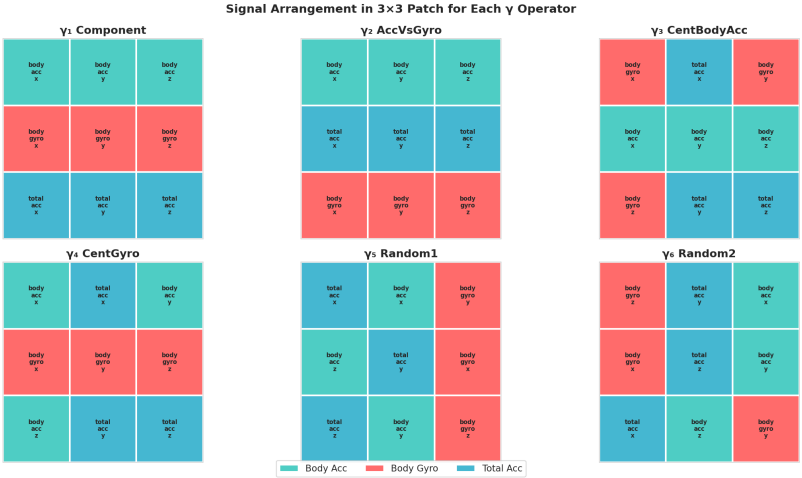


Designing the Six γ Operators

We designed six spatial arrangements (the γ operators) to test different hypotheses about signal adjacency within the 3×3 patch:

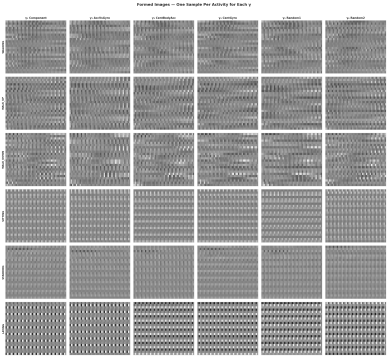
Operator	**Design Principle**
γ_1 (Component-wise)	Each row = one sensor type (highly physically interpretable).
γ_2 (Accel vs. Gyro)	Translational sensors grouped together, rotational separate.
γ_3 (CentBodyAcc)	Body acceleration centered, flanked by gyro and total acc.
γ_4 (CentGyro)	Gyroscope centered, acceleration surrounding.
γ_5 (Cross-modal 1)	Mixes sensor types per row to maximize cross-modal interaction.
γ_6 (Cross-modal 2)	Alternative random mix for robustness checking.

Visualizing the γ Arrangements



Sample Formed Images

- **Dynamic activities:** Periodic horizontal banding from the gait cycle.
- **Static activities (Sitting/Standing):** Uniform horizontal heatmaps.
- **Laying:** Distinct high-contrast static patterns due to gravity shift.



CNN Architectures (1/2): The Baselines

We evaluated 4 architectures (24 experiments total) to separate image-arrangement effects from model-capacity effects.

1. LeNet-5 (Adapted Baseline) — 609,542 Params

- **Exact stack:** Conv(32, 3×3) $\rightarrow$ MaxPool(2) $\rightarrow$ Conv(64, 3×3) $\rightarrow$ MaxPool(2).
- **Classifier head:** Flatten $\rightarrow$ Dense(128) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(6, softmax).
- Simple reference baseline used to track gains from deeper designs.

2. VGG-Style — 763,878 Params

- **Exact stack:** [Conv32 $\rightarrow$ BN $\rightarrow$ Conv32 $\rightarrow$ BN $\rightarrow$ MaxPool $\rightarrow$ Dropout(0.25)]
- then [Conv64 $\rightarrow$ BN $\rightarrow$ Conv64 $\rightarrow$ BN $\rightarrow$ MaxPool $\rightarrow$ Dropout(0.25)]
- then [Conv128 $\rightarrow$ BN $\rightarrow$ MaxPool $\rightarrow$ Dropout(0.25)] $\rightarrow$ Flatten $\rightarrow$ Dense256 $\rightarrow$ Dropout(0.4) $\rightarrow$ Dense128 $\rightarrow$ Dense6.

CNN Architectures (2/2): Modern & Efficient

3. ResNet-Style — 326,086 Params

- **Exact stack:** Conv32 → BN → Residual Block(32, stride 1)
- → Residual Block(64, stride 2) → Residual Block(128, stride 2)
- → GAP → Dense128 → Dropout(0.3) → Dense6.

4. Lightweight Model — 39,334 Params

- **Exact stack:** Conv32 → BN → SepConv64 → BN → MaxPool
- → SepConv128 → BN → MaxPool → SepConv128 → BN
- → GAP → Dense64 → Dropout(0.3) → Dense6.
- Depthwise-separable design gives $\approx 8.3\times$ fewer parameters than ResNet.

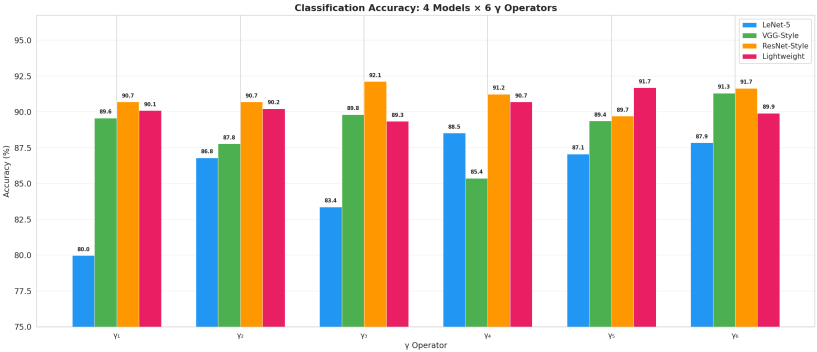
Shared Training Protocol: Adam (10^{-3}), batch size 32, max 50 epochs; EarlyStopping (patience 8) + ReduceLROnPlateau (factor 0.5, patience 4).

Overall Results: Best Configurations

Model	Best γ	Accuracy (%)	F1 Macro (%)
LeNet-5	γ_4 CentGyro	88.53	88.40
VGG-Style	γ_6 Random2	91.31	91.30
ResNet-Style	γ_3 CentBodyAcc	**92.13**	**92.12**
Lightweight	γ_5 Random1	91.69	91.68

- **Top performer:** ResNet-Style paired with γ_3 achieved 92.13% accuracy.
- Exceeds LeNet-5 baseline, proving architectural innovation adds significant value.

How Much Does γ Actually Matter?



- Model choice matters roughly **twice as much** as the γ arrangement ($\sim$ 5.4 pp vs $\sim$ 2.6 pp spread).
- However, no single γ is universally best. The optimal arrangement is architecture-dependent.

Physical vs. Cross-Modal Arrangements

Why γ_3 for ResNet?

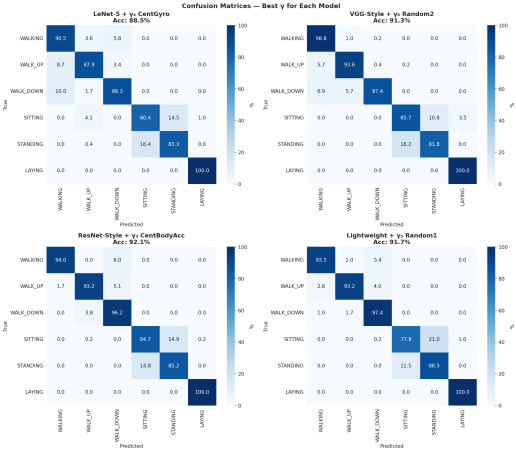
Body acceleration is highly discriminative. Placing it in the center row (γ_3) ensures every 3×3 kernel overlaps with it. ResNet's skip connections propagate gradients efficiently to these first-layer filters.

Why Random/Cross-Modal (γ_5, γ_6) for VGG & Lightweight?

Without skip connections, gradient flow is strictly sequential. Cross-modal arrangements force first-layer kernels to see inter-sensor correlations (e.g., "accel spiked *and* gyro spiked") immediately within their receptive field.

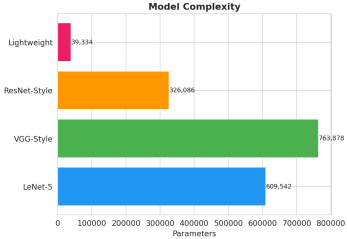
Deep Dive: The Sitting vs. Standing Confusion

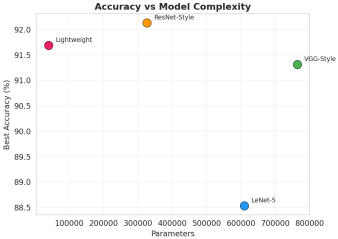
- LAYING is perfectly classified (100%).
- Walking variants are well captured (87-98%).
- **Achilles Heel:** SITTING vs. STANDING.
- Both produce near-identical inertial signals (body is stationary, gravity acts along the same axis). This is a sensor-physics limit, not a model capacity limit.



Model Complexity vs. Deployment Viability

- VGG underperforms despite being the largest (images are too small for deep stacks).
- **Lightweight Model:** Achieves 91.7% with only 39K parameters!
- $\sim 8\times$ smaller than ResNet, proving this pipeline is highly efficient for edge-device (smartphone) deployment.





Avenues for Improvement

1. **Data Augmentation:** Time warping, jitter, and magnitude scaling applied *before* image formation.
2. **Multi-channel (RGB) Images:** Use 3 different γ operators as R, G, B channels simultaneously.
3. **Frequency Domain:** Augment signals with FFT magnitudes for richer images.
4. **Learnable γ:** Replace the fixed permutation with a differentiable attention layer trained end-to-end.
5. **Hierarchical Classification:** Separate dynamic vs. static activities first, then sub-classify to fix the Sitting/Standing bottleneck.

Conclusion

- **Successful Transfer:** Image formation theory successfully adapts from photovoltaic monitoring to Human Activity Recognition (92.13% accuracy).
- **Design Map:** Architecture contributes more to success than signal arrangement, but both are critical.
- **Inter-sensor Correlations:** Cross-modal signal mixing is surprisingly competitive for models lacking skip connections.
- **Efficiency:** Converting time-series to images allows ultra-lightweight models (39K params) to achieve near state-of-the-art performance.

Thank You!